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The Editor
Applied Energy

Dear Editor,

We submit the manuscript *"Training budget and fragmentation, more than heterogeneity, drive the accuracy cost of federated household load forecasting"* by Pushkal Gupta, Vanka Kaushik and Kalaavathi B for consideration as a Research Article in *Applied Energy*.

Federated learning is increasingly proposed as the privacy-preserving way to forecast household electricity demand from smart-meter data. Two of its premises are rarely measured: that its accuracy loss is caused by heterogeneity between households, and that it saves bandwidth compared with collecting the data. Using 100 households from the public Low Carbon London trial, we hold the homes, the test set and the model fixed, change one factor at a time, and count every byte exchanged. The main findings, all for one-step forecasts, are:

- The gap to centralised training grows with the number of clients the same homes are split over, from 1.8 % with five clients to 10.0 % with one client per home after 30 rounds. A control that deals training windows rather than homes, so that clients do not differ at all, keeps 71–96 % of the gap with 20 or more clients in three cohorts.
- Much of this cost is training budget: run to a plateau, at up to 4.7 GB of traffic per run, the gap shrinks to 1.8–5.4 % for homes and 1.3–3.5 % for identical clients, so a fragmentation cost remains without any heterogeneity. A server-side adaptive optimiser recovers part of the budget cost; client-side optimiser changes do not.
- With 20 or more clients, federated training of a small LSTM moves more bytes than the raw half-hourly readings in any encoding, so the case for federation is governance, not bandwidth.
- Local fine-tuning reduces, but does not remove, the share of households that federation leaves worse off; with one home per client, fine-tuned federated models still lose to local ones. Calendar inputs and proximal regularisation do not help.

These results give distribution system operators, suppliers and aggregators concrete guidance: federate a few aggregated clients rather than individual meters, budget rounds by local steps, fine-tune locally and check errors per household. All code, configurations and per-run results are provided, and every number in the manuscript is generated from them.

This manuscript is original, has not been published, and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission. The authors declare no competing interests. The use of AI-assisted tools in preparing the software and the text is disclosed in the manuscript.

Thank you for considering our work.

Yours sincerely,

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